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Title

Reference-guided expiMap prioritizes reproducible tissue-specific pathway shifts in mouse spaceflight transcriptomes

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Abstract

Spaceflight affects multiple organs, but differences among missions can obscure biological responses that recur across studies. We asked whether expiMap, a pathway-constrained variational autoencoder, could identify reproducible biological programs in mouse spaceflight transcriptomes. NASA Open Science Data Repository bulk RNA-sequencing samples were mapped to tissue-matched, non-spaceflight ARCHS4 references trained on approximately 2,000 highly variable genes connected to current mouse Reactome pathways. Flight-ground pathway shifts were summarized with equal mission-project weight and evaluated using conventional gene-set enrichment, held-out-project prediction, three complete reference-query trainings, cell-composition sensitivity analyses, and member-gene review. Four tissues produced reproducible patterns. In skin, flight was associated with lower chromatin regulation, DNA repair, Hedgehog signaling, sphingolipid metabolism, and cell-junction programs, consistent with reduced tissue maintenance and barrier coordination. Thymus showed lower DNA-repair and cytoskeletal programs, together with a model-specific lower lymphoid-stromal interaction score. Liver showed lower MHC class II antigen-presentation and T-cell receptor scores despite heterogeneous metabolic responses. Spleen showed coordinated lower T-cell receptor signaling, neutrophil degranulation, and C-type lectin receptor signaling across five unconfounded projects and three trainings; all three programs had preranked GSEA false discovery rates below 0.05. Reference-guided pathway modeling therefore recovered established tissue responses while identifying additional regulatory, structural, and innate-immune hypotheses. These bulk-tissue associations require independent cell-resolved and functional validation.
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